

SRIRAM VENKAT

+91-6305477216

@ sriramnbv26@gmail.com

ram-s-venkat

sriram-2601

Portfolio

PROFESSIONAL SUMMARY

Motivated Computer Science graduate with hands-on experience in software development, Python programming, cloud computing, APIs, Docker, and full-stack applications. Strong understanding of Data Structures, Software Development Life Cycle (SDLC), Linux environments, networking fundamentals, and database concepts. Familiar with AWS cloud services and cloud-native development. Seeking to contribute to software development, automation, testing, debugging, and infrastructure engineering initiatives as a Software Engineer Apprentice.

TECHNICAL SKILLS

Programming: Python, Java, JavaScript, TypeScript, Object-Oriented Programming (OOP)

Software Engineering: Data Structures & Algorithms, SDLC, Debugging, Testing (Pytest), SDE Fundamentals

Web & APIs: React, Next.js, Node.js, Express, React Flow, Socket.IO, REST APIs

Cloud & DevOps: AWS (EC2, IAM, VPC, Lambda, S3, CloudWatch, ECR), Docker, Git, GitHub

Databases & Queues: SQL, MongoDB, Redis, BullMQ

Systems & Networking: Linux, TCP/IP, HTTP/HTTPS, DNS, IP Addressing, Subnetting, NAT Gateway

AI/ML & Security: PyTorch, FAISS, Sentence-Transformers, Vector Search, OWASP VAPT, NumPy, Pandas

PROJECTS

Serverless Machine Learning Inference via Model Slicing | [Live Link](#) | [GitHub](#) *Oct 2025*

- Designed a distributed split-computing architecture partitioning **MobileNetV3** into 5 sequential slices across edge devices and **AWS Lambda**, bypassing the 250MB serverless deployment limit.
- Reduced peak inference memory consumption from **3,000MB to 600MB** (80% reduction) with zero accuracy degradation using **Amazon S3** tensor streaming and **AWS Step Functions**.
- Containerized microservices using **Docker** and automated cloud infrastructure provisioning with **AWS ECR, IAM, and EC2**.

Agentflow_AI – Agentic Workflow Automation Platform | [Live Link](#) | [GitHub](#) *Aug 2026*

- Engineered an autonomous multi-agent orchestration platform converting natural language prompts into executable visual DAG workflows using **Next.js, Express, and React Flow**.
- Architected a 5-agent execution loop (Planner, Executor, Validator, Recovery, Monitor) with automated retry backoffs, asynchronous task queuing via **BullMQ/Redis**, and **MongoDB**.
- Integrated real-time execution trace streaming using **Socket.IO** and secured third-party OAuth credential tokens (Slack, Gmail, Discord) with **AES-256-CBC** encryption at rest.

OmniSupport AI – Multi-Brand Enterprise Support Agent | [Live Link](#) | [GitHub](#) *Sep 2026*

- Developed an evidence-grounded support AI trained on **106,860+ conversations** across 108 enterprise brands, achieving **100% auto-handling precision** across a 200-case golden benchmark.
- Built a hybrid retrieval pipeline using **FAISS** dense vector indexing and lexical reranking, delivering sub-**48ms CPU inference** without requiring dedicated GPU infrastructure.
- Hardened application resilience against **31 OWASP VAPT security vectors** with deterministic risk gating to eliminate hallucinations on credentials and exploits; validated via **64 automated pytest tests**.

EDUCATION

Keshav Memorial Institute of Technology, Hyderabad

Nov 2022 – May 2026

Bachelor of Technology in Computer Science Engineering (Data Science)

Sri Aadya Junior College

2020 – 2022

Intermediate (MPC) – CGPA: 9.30/10

Janapriya High School

2020

Secondary School of Education – CGPA: 10.0/10.0

CERTIFICATIONS & ACHIEVEMENTS

- Anthropic Certificate – Model Context Protocol: Advanced Topics** | [Link](#)
- Deep Learning in Speech Processing – Guest Lecture Certificate** | [Link](#)
- Generative AI Mastery Workshop – NextWave Certificate of Achievement** | [Link](#)
- Project Expo Finalist – KMIT Project Expo (Dec 2025)** | [Link](#)